MILITARY SPECIFICAT10N

PERISCOPES: XM30 AND XM30C

1. SCOPE

1.1 This specification covers two types of monocular periscopes used as sighting devices for the main armament and coaxial machine gun of the Flame Thrower Tank M67A1.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, forma part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-O-13830 —Optical Components for Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of.

MIL-F-13926 —Fire Control Materiel: General Specification Governing the Manufacture and Inspection of.

MIL-P-14232 —Parts, Equipment and Tools for Ordnance Materiel; Packaging of.

MIL-F-14619 —Fire Control Parts Equipment and Tools; Packaging of.

MIL-I-45208 —Inspection Requirements: General Specification for.

STANDARDS

MILITARY

MIL-STD-105 -Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-130 —Identification Marking of U.S. Military Property.

DRAWINGS

ORDNANCE CORPS

F8634463 —Periscope, XM30 and XM30C

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should he obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Initial production sample. Unless otherwise directed by contracting officer, an initial production sample shall be required. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as will be used in regular production.

3.2 Materials. Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.3 Construction. The periscope shall be manufactured in accordance with Drawing F8634463, "Periscope XM30 and XM30C" and drawings pertaining thereto. The requirements of this specification are detailed only to

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the extent considered necessary to obtain the desired mechanical and optical characteristics, and performance to assure inherent quality and performance capabilities.

3.4 General specifications. The periscope shall meet the following requirements of Specification MIL-O-13830 and MIL-F-13926:

3.4.1 MIL-O-13830

(a) Cleanliness
(b) Illuminated retitles

3.4.2 MIL-F-13926

(a) Order of precedence
(b) Protective finished surfaces
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.5 Environmental.

3.5.1 *Storage temperature.* The periscope shall show no evidence of cement separation, glass breakage or other physical failure, and shall meet all requirements of this specification at standard ambient temperature (60° to 90° Fahrenheit) after having been exposed and thermally stabilized at ambient temperatures of minus 80° Fahrenheit and plus 160° Fahrenheit.

3.5.2 Operating *temperatures.* The periscope shall meet the applicable requirements of 3.6.9 while exposed and thermally stabilized at ambient temperatures of minus 40° Fahrenheit and plus 150° Fahrenheit.

3.5.3 *Vibration.* There shall be no visible damage or loosening of assembled parts as a result of vibration in a vertical plane at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds with the input coupling immobilized and the periscope mounted in its normal operating position. Subsequent to vibration, the periscope shall meet all of the requirements of 3.5.4 to 3.6.10 inclusive.

3.5.4 Cleanlimess. The periscope shall meet the cleanliness requirements of Specification MIL-O-13830, and in addition there shall be no more than two particles of dirt visible on the surface of the reticle. Each particle shall not be in excess of an area equivalent to a size ten dig. There shall be no other dirt which interferes with observation or adversely affects image quality.

3.5.5 *Sealing.* The periscope shall show no evidence of leakage when subjected to an internal pressure of 5.0 psig for a period of five minutes. The interior atmosphere of the accepted periscope shall contain nitrogen having a dew point at least as low as minus 25° Fahrenheit at a pressure of between 0.1 and 0.25 psig.

3.6 Performance.

3.6.1 *Collimation.*

3.6.1.1 *Machine gun reticle.* With a line of sight established perpendicular to surface "D" in elevation (Drawing F8634463) and parallel to a line passing through the center of the locating pin and the vertical axis of the periscope within 1 mil, and when the periscope line of sight is parallel to its horizontal mounting surface within 10 mils, the boresight aiming cross of the machine gun reticle shall be set in coincidence (using boresight knobs) with the established line of sight, and the boresight scales shall be slipped to the normal "(10)" reading. From this position the azimuth and elevation boresight knobs shall provide at least 30 mils movement on each side of the normal "(10)" setting in elevation and deflection,

3.6.1.2 *Flame gun reticle.* The position of the line of sight through the 100 meter boresight point on the flame gun reticle in relation to the boresight cross of the machine gun reticle shall be 5 mils to the left and 5 mils below for Periscope XM30 and 5 mils to the left and 135 mils below for Periscope XM30C. From the established position the screw adjustment for the flame gun reticle shall provide at least 30 mils movement in each direction in elevation and deflection.

3.6.2 *Parallax.* Parallax at the optical axis shall not exceed 0.3 mil at the machine gun reticle when viewing a target between 25 yards and infinity, Parallax between the ma-

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chine gun reticle and the flame gun reticle shall not exceed 0.2 mil.

3.6.3 *Eyepiece focus.* The eyepiece focus shall be fixed between the limits of minus 0.75 and minus 1.0 diopter.

3.6.4 *Resolution.* Resolution on the optical axis shall be 30 seconds of arc or less.

3.6.5 *Elevation trucking accuracy.* The angle between the line of sight through the machine gun reticle boresight cross and surface "D" Drawing F8634463 (90° nominal) shall remain constant within plus or minus 1.0 mil, excluding backlash, at any setting of the coupling between 12 degrees depression and 45 degrees elevation, and within plus or minus 1.5 mils at any setting above 45 degrees elevation up to and including 60 degrees and below 12 degrees depression down to and including 15 degrees. The elevation or depression angle is defined as the angle between the line of sight and the horizontal mounting surface of the instrument.

3.6.5.1 Backlash. Backlash between the optial line of sight and surface "D" shall not exceed 0.2 mil at any setting of the coupling,

3.6.6 *Elevation excursion limits.* Actuation of the lever (Drawing A7633265) of the coupling shall cause an angular change in the line of sight from 15 degrees depression to 60 degrees elevation, a total excursion of 75 degrees. There shall be a full 48 degree field of view with no evidence of cutoff at any setting.

3.6.7 *Alinement.* With the periscope mounting surface perpendicular to a plumb line, the following conditions shall be met:

3.6.7.1 *Plumb travel.* After establishing coincidence with a plumb line at 60 degrees elevation, the line of sight shall not deviate from the plumb line by more than 1.0 mil, nor shall the total lateral spread exceed 1.0 mil, when the line of sight is depressed through the total excursion of 75 degrees.

3.6.7.2 *Parallelism of reticle and image.* With the mounting surface of the periscope horizontal, the 200 meter range line of the machine gun reticle shall be parallel to the image of a horizontal line within 30 minutes of are

3.6.7.3 *Parallelism of flame gun and machine gun reticle.* The flame gun reticle shall be parallel to the machine gun reticle within one (1) degree.

3.6.7.4 *Image tilt.* The image of a vertical target shall be vertical within one degree. (One degree tilt is equivalent to the angle subtended by 14.3 roils at the edge of the field when measured over the full field of view.)

3.6.8 *Reticle illumination.* When illuminated by an Instrument Light, T25 or equal, the reticle markings shall appear clearly defined when observed in a darkened room.

3.6.9 *Torque.* The torque required to rotate the coupling and boresight knobs shall not exceed 7 pound inches at standard ambient temperature (60° to 90° Fahrenheit) and when measured at temperatures specfied in 3.5.2, shall not have changed from the value obtained at standard ambient temperature by more than five pound inches.

Caution: At no time shall the coupling be allowed to snap (spring return) against the travel stop.

3.6.10 *Boresight mechanism backlash.* Backlash in the boresight mechanism shall not exceed 0.1 mil and reticle shift due to movement of either knob shall not exceed 0.2 mil in any direction.

3.7 Identification marking. Unless otherwise specified, identification marking shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibilities. The supplier is responsible for the performance of all inspection *requirements* as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to

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perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General requirements. Inspection requirements and procedures shall be in accordance with Specification MIL-I-45208.

4.3 Initial Production sample. Prior to production the contractor shall submit the initial production sample in accordance with 3.1 for inspection by the contractor under the surveillance of the Government. The inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The sample shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.3.1 *Production sample failure.* Should the production sample fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. Where corrections in the process have been made, a new production sample shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production sample is submitted and corrective actions have been taken to eliminate deficiencies.

4.4 Lot formation. Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.1 *Defective units.* Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.5 Material. Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection of components. All positional and functional characteristics shown on drawings, *or* contained in notes on the drawings or referenced specifications, which indicate precision requirements and have a bearing on the functional accuracy and performance of periscopes shall be inspected in accordance with the written inspection system required by Specification MIL-L-45208. These characteristics are defined as requirements effecting concentricity, squareness, parallelism, microfinishes 64 RMS or better, threads, identification marking and performance.

4.6.1 *Sampling inspection.* Where quality control has been established on in-process manufacture, sampling inspection of components may be used in lieu of 100 percent inspection and shall be in accordance with table I.

TABLE I. *Sampling plan*

Lot size inclusive	Sample size	Accept lot	Reject lot
0–25	Use 100% inspection		
20–35	20	0	1
36–50	25	0	1
51–150	30	0	1
151–300	30	1	2
301–over	60	2	3

4.6.2 *Disposition of nonconforming product.* Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of nonconforming components shall be in accordance with Standard MIL-STD-105.

4.7 Inspection equipment. Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.7.1 *Accuracy requirements.* Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance except as specified in 4.7.2. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

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4.7.2 *Inspection equipment design equipments.*

4.7.2.1 Device to vibrate the periscope. The device shall be a type capable of producing vertical motion at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch excursion). The device shall be equipped with an adapter which will permit the mounting of the periscope in its normal operating position as specified in 3.5.3. The base of the adapter shall be secured firmly to the device.

4.7.2.2 *Device to test sealing of the periscope. To* determine compliance with 3.5.5, the device shall be capable of charging the interior of the periscope with dry nitrogen to an internal pressure of at least 5.0 psig (pounds per square inch gage). This device shall be equipped with a pressure gage capable of registering the internal pressure up to at least 5.0 psig, subdivided into increments of 0.2 psig. The accuracy of the pressure gage shall be within ten percent at any point on the scale.

4.7.2.3 *Device to test accuracy of the periscope.* The device shall maintain the horizontal mounting surface of the periscope in a horizontal plane. The device shall provide a means of positioning surface "D" perpendicular to the periscope horizontal mounting surface. The device shall also provide for measured angular positioning of surface "D" with reference to the horizontal mounting surface.

4.7.2.3.1 *Target to measure accuracy of the periscope.* An appropriate target with acceptance and rejection limitation markings or pips marked thereon shall be utilized for testing the accuracy of the periscope. The accuracy requirements for positioning the limitation marks or pips shall be as follows:

(a) Collimation, machine gun reticle, within 0.25 mil
(b) Collimation, flame gun reticle, within 0.25 mil
(c) Elevation tracking accuracy, within 0.1 mil
(d) Backlash, elevation tracking, within 0.1 mil
(e) Excursion limits, within 10 minutes
(f) Field of view, within 10 minutes
(g) Plumb travel, within 0.1 mil
(h) Parallelism of reticle and image, within 3.0 minutes
(i) Parallelism of retitles, within 10 minutes
(j) Image tilt, vertical line plumb

4.8 Acceptance inspection.

4.8.1 *Classification of defects.* The classification of defects in table II and III shall constitute the minimum inspection requirernents to be performed prior to acceptance or rejection by lot.

4.8.2 *Disposition of rejection lots.* Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrected action taken, the lot shall be resubmitted for acceptance inspection as outlined in Standard MIL-STD-105.

TABLE II. ---*Classification* of *defects*

Use Inspection Level L7, in table III-B, with Sampling Plan table IV-A of Standard MIL-STD-105.

CRITICAL: None defined.

101. Coupling, excessive torque, ...	3.5.2	4.9.2	Standard Measuring. Equipment (SME).
102. Boresight knobs, excessive torque	3.6.2	4.9.2	SME.
108. Sealing failure	3.6.6	4.9.6	Special Testing Equipment (STE).
104. Recticle defective or inadequate illumination. . .	3.6.8	4.9.9	Visual.

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MINOR: None defined.

Note. The tests for characteristics 101 and 102 shall be conducted at minus 40° and plus 160° Fahrenheit, *respectively.* The test for characteristics 103 and 104 shall be conducted at standard ambient *temperature* subsequent to the operability tests at the temperature extremes, Tests shall be conducted in accordance with the quality assurance paragraph (QA par.) listed.

TABLE III. *Classification of defects*
Use Inspection Level II, of table III-A, with Sampling Plan table IV-A of Standard MIL-STD-105

CRITICAL: None defined.
MAJOR: 1.5%

		Requirement paragraph	*QA paragraph*	*Test method*
101.	Physical failure, post vibration	3.6.3	4.9.8	Visual-Tactile
102.	Cleanliness	3.6.4	4.9.4	Visual
103,	Collimation error, machine gun reticle	8.6.1.1	4.9.6.1	Special Testing Equipment (STE) visual
104.	Collimation error, flame gun reticle	3.6.1.2	4.9.6.2	STE-Visual
105.	Parallax excessive	3.6.2	4.9.6.9	STE-Visual
106.	Elevation tracking	3.6.5	4.9.6.4	STE-Visual
107.	Backlash excessive, elevation tracking	3.6.5.1	4.9.6.6	STE-Visual
108.	Elevation excursion limits inadequate	3.6.6	4.9.6.6	STE-Visual
109.	Field of view inadequate	3.6.6	4.9.6.6	STE-Visual
110.	Plumb travel *error*	3.6.7.1	4.9.6.7	STE-Visual
111.	Machine gun reticle cant excessive	3.6.7.2	4,9.6.8	STE-Visual
112.	Flame gun reticle cant excessive	3.6.7.3	4.9.6.9	STE-Visual
ll3.	Image tilt excessive	3.6.7.4	4.9.6.10	STE-Visual
114.	Eyepiece focus error	3.6.3	4.9.7	Standard Measuring Equipment (SME) Visual
115.	Resolution inadequate	3.6.4	4.9.8	STE-Visual
116.	Coupling, excessive *torque*	3.6.9	4.9.10	SME
117.	Boresight knobs, excessive torque	3.6.9	4.9.10	SME
118.	Backlash excessive, boresight mechanism	3.6.10	4.9.11	STE-Visual
119.	Identification markings incorrect	3.7	4.9.12	Visual

Note. The above tests shall be conducted at standard ambient temperature in accordance with the quality assurance paragraph (QA par.) listed

4.9 Test methods and procedures.

4.9.1 *Storage temperature.* The temperature cycling for the physical endurance test of the periscope at minus 80° and plus 160° Fahrenheit shall be conducted in accordance with Procedure II of Specification MIL-F-13926. Upon return to standard ambient temperature (plus 60° to plus 90° Fahrenheit), the periscope shall be subjected to a visual and tactile examination to determine compliance with 3.5.1 and the periscope shall show no evidence of physical failure. The periscope shall then be subjected to the tests specified in 4.9.3 through 4.9.11 inclusive and shall meet the requirements specified. The storage temperature test shall be performed on the production sample only.

4.9.2 *Operating temperatures.* The temperature cycling for the operability test of the periscope at the specified temperature extremes in 3.5.2 shall be conducted in accordance with Procedure I of Specification MIL-F-13926, except that the temperature shall not be reduced beyond minus 40° Fahrenheit or raised beyond plus 150° Fahrenheit. When the periscope is thermally stabilized at each temperature extreme, it shall be subjected to the tests in 4.9.10. The controls of the periscope shall meet the specified torque requirements for the indicated operating temperatures.

4.9.3 *Vibration.* The periscope shall be mounted in special testing equipment and vibrated in accordance with the requirements specified in 3.5.3. During the test, the input coupling shall be immobilized in a position that shall prevent it from contacting the travel stop pin. Subsequent to vibration, the

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periscope shall be subjected to a visual-tactile examination for damage. The periscope shall then be subjected to the tests specified in 4.9.4 through 4.9.11 inclusive, and shall meet the requirements specified.

4.9.4 *Cleanliness.* The periscope shall be inspected for cleanliness in accordance with Specification MIL-O-13830 and shall meet the requirements specified in 3.5.4 The mean diameter of a size 10 dig is equal to 0.10 millimeter.

4.9.5 *Sealing.* The periscope shall be internally pressurized to the psig value specified for the required time in compliance with 3.5.5. During this time period the interior pressure of the periscope shall meet the requirements specified. Internal pressure shall be measured with special testing equipment. Upon satisfactory completion of the sealing tests, the periscope shall be flushed with dry nitrogen. The flushing shall continue until the gas bled from the periscope has a dewpoint at least as low as minus 25° Fahrenheit at the pressure specified. The periscope shall then be sealed. This is a post environmental test that shall be performed at plus 60° to plus 90° Fahrenheit.

4.9.6 *Accuracy tests.* The following tests shall be conducted with the periscope positioned on special testing equipment that conforms to the design requirements of 4.7.2. Surface "D" of the periscope coupling shall be positioned perpendicular to the periscope horizontal mounting surfaces unless otherwise specified. The target shall lie in a plane that is perpendicular to a line passing through the mechanical axis of Pin 8614756 and the vertical axis of the periscope. The distance between the target and periscope shall be equal to or greater than the minimum parallax distance specified in 3.6.2.

4.9.6.1 *Collimation machine gun reticle.* Using the boresight knobs, position the boresight point of the machine gun reticle coincident to the intersection of the vertical and horizontal target line image. Slip the knob scales to the position as required in 3.6.1.1. Rotate the azimuth and elevation boresight knobs from the established normal position to their counterclockwise and clockwise stops. This rotation of the knobs shall move the horizontal and vertical reticle line the required number of mils each side of the established normal position, as measured in the target to determine compliance with 3.6.1.1. Upon completion of this test return the boresight knobs to their established normal position.

4.9.6.2 *Collimation flame gun reticle.* Position the boresight point of the machine gun reticle as specified in 4.9.6.1. Using the screw adjustments position the boresight point of the flame gun reticle the required number of mils in azimuth and elevation as measured on the target. Record position of boresight point on target to determine compliance with 3.6.1.2. The screw adjustments shall permit the frame gun reticle boresight point to be moved the specified number of mils each side of the recorded position as measured on the target. Upon completion of this test return the recticle boresight to the recorded position.

4.9.6.3 *Parallax.* Parallax shall be measured at the optical axis. Any movement of the observer's head shall not cause the machine gun reticle boresight mark to displace more than the allowable tolerance specified in 3.6.2 when viewing a target between 25 yards and infinity. Parallax between the two retitles at the boresight point of the flame gun reticle shall not exceed the tolerance specified.

4.9.6.4 *Elevation tracking accuracy.* Position surface "D" of the input coupling perpendicular to the periscope mounting surface with backlash removed for direction of travel, Any measured change in the angular position of surface "D" of the input coupling shall cause the same measured angular change in the line of sight within the tolerance specified. Elevation accuracy shall be measured at a minimum of four positions in elevation and two positions in depression starting with the line of sight horizontal when determining compliance with 3.6.5.

4.9.6.5 *Backlash.* This test will be conducted in conjunction with the test specified in 4.9.6.4 to determine compliance with 8.6.5.1. Backlash shall be measured at each setting by taking successive readings in reverse direc-

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tions and shall not exceed the tolerance specified.

4.9.6.6 *Elevation excursion limits.* The field of view and excursion limits shall be measured while actuating the coupling causing an angular change in the line of sight. The limits of excursion and width of the field of view as measured on the target shall not be less than the amount specified in 3.6.6.

4.9.6.7 *Plumb travel.* The elevating mechanism shall be actuated until 60° elevation has been attained. At this point of elevation, coincidence with a plumbline target shall be established with the boresight cross of the machine gun reticle. The line of sight shall be depressed until maximum depression has been reached. Throughout the complete depression excursion, the line of sight shall not deviate in excess of the allowable tolerances of 3.6.7.1, as measured on the plumb line target.

4.9.6.8 *Parallelism of reticle and image.* When one extremity of the horizontal reticle line is coincident to the horizontal target line image, the reticle line shall be parallel to the target line image within the tolerance specified in 3.6.7.2 as measured on the target.

4.9.6.9 Parallelism of flame gun and machine gun reticle. This test will be conducted in conjunction with the test in 4.9.6.8. Establish coincidence of the left extremities of a flame gun and machine gun reticle line. The flame gun reticle shall be parallel to the machine gun reticle within the tolerance specified in 3.6.7.3, as measured on the target.

4.9.6.10 *Image tilt.* This test shall be performed with the aid of an auxiliary telescope. The vertical reticle line of the auxiliary telescope shall be adjusted parallel to the image of the vertical target line. When the periscope is positioned in front of the auxiliary telescope, the image of the vertical target line shall be parallel to the image of the vertical reticle line of the auxiliary telescope within the tolerance specified in 3.6.7.4, as measured on the target.

4.9.7 *Eyepiece focus.* Focus shall be measured with the aid of a calibrated dioptometer. The eyepiece of the dioptometer shall be adjusted for best focus of the dioptometer reticle with the diopter measuring scale set at zero. Position the dioptometer at the eyepiece of the periscope. The machine gun and the flame gun retitles shall be in best focus in the center of the field within the tolerance specified in 3.6.3, as measured on the dioptometer scale.

4.9.8 *Resolution.* This test shall be conducted in accordance with Specification MIL O-13830 and the resolving power on the optical axis shall meet the requirements specified in 3.6.4.

4.9.9 *Reticle illumination.* The retitles shall be illuminated with a device consisting of two 11/2 volt batteries and a .35 to .5 amp bulb. The markings of the machine gun and flame gun retitles shall be visually examined against a dark background to determine compliance with 3.6.8. Reticle markings shall be clearly defined. Should the brightness of a reticle defect be greater than the brightness of a reticle line, the defect shall be cause for rejection.

4.9.10 *Torque.* Torque shall be measured with standard measuring equipment. The measured torque required to rotate the coupling and boresight knobs shall not exceed the values specified in 3.6.9 for the temperature indicated.

4.9.11 *Boresight mechanism backlash.* With the periscope mounted as specified in 4.9.6, the boresight knobs shall be checked for backlash. Rotate the boresight knobs, individually, in a clockwise direction to bring a reference point on the machine gun reticle in alignment with a target reference point and note the reading on the boresight knob micrometer. Rotate the knobs in the same direction to their limit, then rotate counterclockwise until realignment of the same reticle reference point and target reference point is obtained. Read the boresight knob micrometer again. Any difference in the readings shall be construed as backlash and shall not exceed the tolerances. During the above procedure note any shift of the reticle in a direction other

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than normal. Reticle shift shall not exceed the allowable tolerance.

4.9.12 *Identication marking.* Markings shall be visually inspected for conformance to Standard MIL-STD-130.

4.10 Packaging, packing and marking. Quality assurance provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232,

4.11 Final acceptance. Final acceptance of a lot of periscopes shall be withheld by the Government inspector until the sample units, representative of that production lot, have satisfactorily passed the applicable requirements of this specification.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with Specification MIL-F-14619/ P8634463. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. The Periscope XM30 and XM30C when installed and adjusted in Flame Thrower Tank M67A1, shall track both the flame gun and the coaxial machine gun from 12 degrees depression to 45 degrees in elevation. Periscope XM30 and XM30C have a 1.5 power optical system and a 48 degree field of view.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Selection of applicable level of preservation, packaging and packing.

(c) Applicable stock number.

(d) Provisions for production sample including date of delivery (see 3.1).

6.3 Definitions.

6.3.1 *Standard measuring equipment (SME).* Standard measuring equipment is defined as the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.3.2 *Special testing equipment (STE).* Special testing equipment is defined as special inspection fixtures, instruments and modified standard measuring equipment.

Notice. When Government drawings, specifications, or other data are used for *any* purpose other than in connection with a definite] y related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government *may* have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodian:
Army—Ord

Preparing activity:
Army—Oral

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